Required Capstone Checkpoint 3.1: Evaluation Infrastructure and Baseline Diagnosis

Directions

Download this worksheet and save a copy to your device before you begin working. Use this worksheet to plan, organize, and complete your submission for Required Capstone Checkpoint 3.1. Type your responses directly into the document, save your completed work, and upload the finished file when you are ready to submit.

Completion Checklist

☒ A clear description of the retrieval system and intended user tasks
☒ Defined and justified evaluation criteria, metrics, and testing approach
☒ Baseline evaluation results supported by relevant examples or evidence
☒ Identification of meaningful system strengths, weaknesses, and likely failure modes
☒ Evidence that the evaluation framework can detect degraded, manipulated, or otherwise problematic outputs
☒ Reflection on potential blind spots or limitations in the evaluation framework

Step 1. System Overview

Describe the baseline retrieval system you developed for your selected capstone scenario.

- Briefly explain what the system retrieves or answers.
- Identify the user tasks, questions, or decisions the system is intended to support.

Brief description of your retrieval system
Response: **Scenario and objective.** I built a retrieval-augmented Wikipedia question-answering system over a local corpus of 2,417 HTML articles. Each question is independent, and the answer model is instructed to use only the context supplied for that question. **Two-stage baseline.** I first ran an oracle-context evaluation in which the full designated source articles were supplied directly, with no retrieval. This estimates answer-generation performance when the intended evidence is available. I then ran the same test set through vector retrieval and vector-heavy hybrid retrieval. Both used Top K = 7 seed chunks and expanded every selected article to its full extracted text. **Controlled comparison.** The vector and hybrid runs used the same 19 questions, grading notes, answer model, judge model, temperature, Chroma index, and context-expansion process. The only intended

© MIT xPRO® 2026. All rights reserved.

change was the retrieval method. The hybrid configuration used BM25/vector weights of 0.35/0.65 and a candidate pool of 10.

Reason for the experiment. My manual Checkpoint 2.1 review suggested that vector search was more useful for this corpus and that BM25 sometimes added lexically related but irrelevant articles. Checkpoint 3.1 tests that observation with a structured, repeated evaluation rather than relying on individual examples.

The detailed measurements and visual comparisons are provided in **ragas_evaluation_report_checkpoint3_1.pdf**.

Intended user tasks or questions

Response:

The system is intended to support trustworthy Wikipedia research and fact finding across several user tasks:

- Precise facts and short quotations from a single article.
- Open-list questions, process explanations, and table lookups.
- Time-bounded filtering, counting, and exhaustive aggregation.
- Comparisons and intersections across multiple articles.
- Connections among people, works, awards, political roles, places, and legal entities.
- Geopolitical timelines, historical identity resolution, arithmetic across tables, uncertainty preservation, and limits on causal inference.

The attached **wikipedia_GT.csv** contains 19 cases classified by category, evaluation role, scope, assumptions, and complexity. The questions range from direct lookups to multi-source reasoning tasks designed to reveal retrieval, context, synthesis, and evaluation failures.

Step 2. Evaluation Design

Define how you will evaluate the baseline system before introducing further optimization or architectural changes.

- Identify the criteria you will use to judge system performance.
- Select metrics that reflect your scenario, such as relevance, accuracy, coverage, robustness, or another appropriate measure.
- Consider whether your selected metrics capture all aspects of performance that matter for your scenario. What important behaviors might not be fully measured?
- Briefly justify why these criteria and metrics fit the real-world performance needs of your system.

© MIT xPRO® 2026. All rights reserved.

Evaluation criteria

Response:

I evaluated five related behaviors:

- **Factual correctness.** Every stated fact, date, value, identity, relationship, and calculation must match the question-specific grading notes.
- **Completeness.** The response must include all required items and requested details, including quotations and uncertainty when applicable.
- **Grounding and instruction compliance.** The response must stay within the supplied evidence, avoid unsupported additions, and respect filters such as dates, official program debuts, or geographic scope.
- **Retrieval coverage.** The retriever should return the designated article or articles that can support the answer.
- **Robustness.** Repeated runs should produce stable outcomes, especially for complex questions.

The grading notes in the ground-truth CSV are authoritative for each question. They define required facts, acceptable variants, partial-credit conditions, and material errors.

Selected metrics

Response:

Answer outcome rates. I report the percentage of PASS, PARTIAL, and FAIL judgments. PASS means the required answer is complete and correct. PARTIAL means the supplied information is correct but incomplete, with no material false addition. FAIL means the answer contains a material error, unsupported claim, invalid addition, or contradiction, even if part of the answer is correct.

Answer quality index. For a compact comparison, PASS = 1.0, PARTIAL = 0.5, and FAIL = 0.0. Raw outcome percentages remain the primary evidence because the weighting is a reporting choice rather than a universal measure.

Designated-source recall. For each question, I divide the number of designated source articles retrieved by the number listed in the ground truth. Full designated-source coverage is the percentage of cases with recall equal to 1.0.

Context size. Mean context characters measure how much expanded article text was sent to the answer model. This is a useful indicator of noise and efficiency, although it is not an exact token or cost measurement.

Stability and task-level behavior. The report compares repeated-run variability and per-question outcomes so that aggregate scores do not hide systematic failures.

Potential metric limitations

© MIT xPRO® 2026. All rights reserved.

Response:

- The answer model and RAGAS judge both used openai/gpt-5.4-mini, which can create correlated blind spots or self-preference.
- The judge receives the response and detailed grading notes, not an independent reading of every source article. A fact omitted from the grading notes may not be checked.
- One discrete label combines correctness, completeness, grounding, and instruction compliance, and the experiment files do not preserve a judge rationale.
- Designated-source recall does not measure retrieval precision and may undervalue a different article that contains equivalent evidence.
- The 19 questions are manually curated and were refined while the framework was developed, so they function as a development set rather than an unseen benchmark.
- Repeated runs revealed nondeterministic judgments, especially on GT07 and GT08. A single run should not be treated as definitive.

Instrumentation limitation. An early .html versus .html.txt filename mismatch produced incorrect recall values. The analysis utility now normalizes those suffixes and recomputes recall, but the incident showed that the measurement code also requires validation.

Brief justification for your criteria and metrics

Response:

For a research-oriented retrieval system, a plausible answer is not sufficient. A user must be able to trust that requested facts are correct, that important items were not silently omitted, and that unsupported conclusions were not added.

The separation between answer quality and retrieval coverage makes the results diagnostic. If oracle context is present but the answer fails, the likely problem is generation, evidence presentation, or judging. If designated sources are missing, retrieval or query handling is a more likely cause. Context size helps determine whether additional recall introduced excessive noise.

This framework therefore supports both real-world trust requirements and concrete development decisions.

Step 3. Testing Approach

Describe the test set and process you used to evaluate the baseline retrieval system.

- Include the number and types of queries or cases used.

© MIT xPRO® 2026. All rights reserved.

- Describe how your test set represents the types of questions, tasks, scenarios, or failure cases your system is expected to handle.
- Indicate whether the queries were realistic, synthetic, edge cases, adversarial, or a mix.
- Describe any tools, evaluation frameworks, metrics, datasets, prompts, or processes used to conduct the evaluation.

Description of your test set
Response: The test set contains **19 ground-truth questions** with stable IDs GT01 through GT19. Each row includes a category, question, reference answer, PASS/PARTIAL/FAIL grading notes, designated source articles, scope assumptions, evaluation role, and complexity. The cases cover direct facts and quotations, open lists, table lookup, process explanation, cross-article intersections, temporal aggregation, uncertain estimates, historical identity resolution, people connections, geopolitical timelines, cross-table arithmetic, and cautious interpretation of population and life-expectancy data. The questions are purpose-built but realistic Wikipedia user tasks. Several are deliberate edge cases: exhaustive lists without telling the model how many answers exist, returning Olympic sports that must not be treated as debuts, disputed fatality estimates, historical country names, time-period mismatches, and questions where the supplied evidence cannot establish causation. The set does not yet include prompt-injection or security adversarial cases.

How the evaluation was conducted
Response: **Ground-truth preparation.** I audited the original Checkpoint 2.1 workbook against the local article-text store, removed a duplicate, clarified ambiguous wording, added question-specific grading rules, and expanded the set to 19 cases. **Oracle-context phase.** For each question, the program loaded the complete text of every designated source article, generated an answer, and used a RAGAS DiscreteMetric to assign PASS, PARTIAL, or FAIL. I completed 9 comparable oracle runs. **Retriever phase.** The same questions were evaluated with vector Top K = 7 and hybrid Top K = 7 with BM25/vector weights 0.35/0.65. Retrieved chunks selected unique articles, which were expanded to full text before answering. I completed 5 vector and 5 hybrid runs. **Controlled models.** The answer model and judge were openai/gpt-5.4-mini through OpenRouter at temperature 0.2. Vector retrieval used openai/text-embedding-3-small. Question wording, grading notes, generation prompt, context strategy, and models were held constant across the vector/hybrid comparison. Note: Selecting different model for judgement was prohibited by preset OpenRouter guardrails.

© MIT xPRO® 2026. All rights reserved.

Evidence and analysis. RAGAS saved one CSV per run. The analysis utility excludes eight earlier binary pass/fail runs because they are incompatible with the final three-label rubric, removes duplicate files, normalizes article filenames, recomputes retrieval coverage, and produces the attached PDF.

The final comparison contains 19 repeated runs and 361 scored responses: 171 oracle responses, 95 vector responses, and 95 hybrid responses. These are repeated measurements of 19 questions, not 361 independent test cases.

Step 4. Baseline Results

Present the baseline performance results from your evaluation.

- Include scores or findings across your selected metrics. Note any surprising, inconsistent, or difficult-to-interpret results.
- Provide at least two concrete examples of system outputs.
- Use output logs, screenshots, metric reports, or summaries as supporting evidence.

Summary of baseline evaluation results

Response:

Oracle context - 9 runs, 171 responses. PASS 77.2%, PARTIAL 3.5%, FAIL 19.3%, quality index 0.789. The oracle result is an answer-generation and judge ceiling for this test set, not a guarantee of a correct response.

Vector Top K = 7 - 5 runs, 95 responses. PASS 72.6%, PARTIAL 4.2%, FAIL 23.2%, quality index 0.747, designated-source recall 89.9%, full designated-source coverage 78.9%, and mean context size approximately 127,000 characters.

Hybrid Top K = 7, BM25/vector 0.35/0.65 - 5 runs, 95 responses. PASS 68.4%, PARTIAL 4.2%, FAIL 27.4%, quality index 0.705, designated-source recall 88.2%, full designated-source coverage 73.7%, and mean context size approximately 184,600 characters.

Primary finding. Vector search was the strongest tested end-to-end configuration. Hybrid retrieval sent about 45% more context while producing lower recall and answer quality. This supports my Checkpoint 2.1 observation for this corpus, test set, weighting, and Top K; it does not show that BM25 is universally ineffective.

Important surprise. The oracle quality index was only 0.789. Supplying the intended articles removed retrieval as a failure source but did not eliminate mistakes in exhaustive intersection, table interpretation, temporal aggregation, and uncertainty handling.

© MIT xPRO® 2026. All rights reserved.

Example 1: Query, system output, and result
Response: **GT01 direct fact and quotation.** Vector retrieval returned 100_metres.html and Usain_Bolt.html, including the designated source, for recall 1.0. **Output excerpt.** “The current men's world record in the 100 metres is 9.58 seconds, held by Usain Bolt of Jamaica. It was set on 16 August 2009 at the World Athletics Championships final in Berlin, Germany.” The response also supplied the requested faithful quotation. **Result.** PASS in confident_valiant.csv. This case shows that a precise fact stated in one relevant passage is handled reliably.

Example 2: Query, system output, and result
Response: **GT07 time-bounded aggregation.** The oracle run supplied the complete Best Actor article. **Output excerpt.** “The actor who won most often was Daniel Day-Lewis ... with 2 wins.” The answer correctly listed There Will Be Blood and Lincoln but omitted Sean Penn, who tied with two wins. **Result.** PARTIAL in elated_brin.csv. Every included claim was correct, but the exhaustive answer was incomplete. This example confirms why a binary pass/fail rubric was insufficient.

Example 3: Query, system output, and result, if applicable
Response: **GT10 Winter Olympic program history.** Vector retrieval returned Winter_Olympic_Games.html, giving designated-source recall 1.0. **Output excerpt.** The answer correctly named biathlon, luge, freestyle skiing, short-track speed skating, snowboarding, and ski mountaineering, but also added “Skeleton - 2002.” Skeleton first appeared before 1950 and 2002 was a return, not its debut. **Result.** FAIL in confident_valiant.csv because the rubric treats any invalid addition as a material error. The failure occurred despite full designated-source recall, so the problem was table/timeline interpretation rather than article retrieval.

Step 5. Performance Analysis

Analyze what the baseline results reveal about your system.

© MIT xPRO® 2026. All rights reserved.

- Identify meaningful strengths and weaknesses.
- Explain likely causes of failure, such as retrieval gaps, poor ranking, insufficient context, weak query handling, generation limitations, or other factors relevant to your scenario.
- Connect your analysis to the evidence presented in Step 4.

Key system strengths

Response:

- Precise single-article facts and quotations were reliable. GT01 passed in every included run.
- Vector retrieval achieved the best end-to-end balance: 89.9% designated-source recall, 72.6% pass rate, and substantially less context than hybrid retrieval.
- The framework distinguishes source selection from answer synthesis. This made it possible to identify failures that persisted even with oracle context or full source recall.
- The model handled several multi-hop relationship and geopolitical questions well and passed GT19's India/China calculation and causal-limitation task in all vector and hybrid runs.
- Stable IDs, detailed grading notes, repeated runs, stored CSVs, and the automated report make later configurations directly comparable.

Key system weaknesses

Response:

- Exhaustive cross-article intersection was the hardest task. GT03 failed in every included oracle, vector, and hybrid run even though oracle mode supplied all three ceremony articles.
- Table and timeline semantics remained weak. GT10 failed in every included run, and GT11 failed in all vector and hybrid runs, usually because returning or future sports were misclassified as first appearances.
- Full designated-source recall did not guarantee a correct answer. GT10 is the clearest example.
- The oracle quality index of 0.789 shows that generation and evidence interpretation remain material bottlenecks even after retrieval is removed.
- Hybrid retrieval increased context size without improving quality, indicating that the answer model can be distracted by additional full articles.

Likely failure modes and causes

Response:

- **Retrieval gaps and ranking noise.** Multi-source questions may miss one designated article, while BM25 can elevate literal matches that are not relevant to the requested relationship.

© MIT xPRO® 2026. All rights reserved.

- **Context overload.** Full-article expansion improves coverage, but several selected articles can produce very large prompts and obscure the decisive evidence.
- **Flattened document structure.** HTML-to-text extraction weakens table columns, list hierarchy, bold winner markers, and year ranges needed for Academy Award and Olympic questions.
- **Generation limitations.** The model may omit members of an exhaustive set, confuse first appearances with returns, miscount repeated winners, or contradict evidence already quoted in its answer.
- **Evaluation limitations.** An LLM judge can interpret borderline partial/fail cases inconsistently, especially when grading notes are long or the response contains both correct and misleading material.

Step 6. Evaluation Framework Validation and Reliability

Demonstrate whether your evaluation framework can reliably detect degraded, manipulated, or otherwise problematic outputs.

- Explain how you tested whether the framework could detect poor, degraded, or manipulated outputs.
- Provide evidence from the validation test.
- Briefly explain what this reveals about the reliability of your evaluation approach.
- Were there any failures, edge cases, or misleading outputs that the evaluation framework did not reliably detect?

How you tested the evaluation framework
Response: I added an optional **validate_framework()** smoke test to both evaluation scripts. It uses GT01 and the same grading notes as the main experiment. The test generates a normal answer from the gold article context and compares it with a deliberately manipulated answer. The manipulated version changes every central fact: it claims a 10.20-second record held by Carl Lewis of the United States and says it was set in Paris in 1995. After introducing the three-label rubric, I also checked naturally degraded outputs from repeated experiments. GT07 provides a clean omission case, while GT10 provides a response containing mostly correct data plus one invalid addition. I manually compared these outputs with the question-specific grading notes to confirm that the automated labels matched the intended policy.

© MIT xPRO® 2026. All rights reserved.

Evidence that the framework detected poor, degraded, or manipulated outputs

Response:

- **Correct GT01 response -> PASS.** The answer reported 9.58 seconds, Usain Bolt of Jamaica, Berlin, and 16 August 2009 and included a faithful quotation.
- **Manipulated GT01 response -> FAIL.** The answer substituted 10.20 seconds, Carl Lewis, the United States, Paris, and 1995.
- **Incomplete GT07 response -> PARTIAL.** The response correctly named Daniel Day-Lewis with two wins but omitted tied leader Sean Penn and introduced no false fact.
- **Misleading GT10 response -> FAIL.** The response contained six correct sport/year pairs but incorrectly added Skeleton - 2002. The strict rubric did not allow the correct majority to conceal the false addition.

Brief interpretation of the validation results

Response:

The framework reliably detected the deliberately false GT01 answer and distinguished a clean omission from a materially wrong addition. This is important because incomplete information and false information create different risks for a research user.

The validation also showed that automated evaluation is evidence, not an unquestionable ground truth. Earlier binary judging occasionally accepted incomplete or contradictory answers, and repeated three-label runs still varied on complex cases. I therefore use repeated results, per-question inspection, and manual spot checks rather than relying on one aggregate score.

Potential blind spots in the evaluation framework

Response:

- The answerer and judge use the same model, so they may share reasoning errors or prefer similar wording.
- The judge is not given an independent source-context check. It depends on the grading notes being complete and unambiguous.
- PASS/PARTIAL/FAIL collapses several dimensions into one label and does not explain which fact caused a failure.
- The framework may miss a fabricated but plausible detail that is not explicitly covered by the grading notes.
- Designated-source recall does not show whether the remaining retrieved articles were irrelevant, nor does it recognize every possible alternative source.

© MIT xPRO® 2026. All rights reserved.

- The test set was refined after observing system behavior. Without a held-out set, improvements may overfit these 19 questions.
- Latency, API cost, token count, citation fidelity, retrieval precision, and security behavior are not directly measured.

Step 7. Reflection and Next Steps

Reflect on the limitations of your current evaluation approach and how you would refine it before using it to guide system optimization.

- Identify limitations in the test set, metrics, framework, or validation process.
- Propose specific improvements you would make next.
- Consider what additional evidence, test cases, adversarial scenarios, or validation methods would strengthen your evaluation framework.

Limitations of your current evaluation approach
Response: • The evaluation contains only 19 manually designed questions from one static Wikipedia snapshot, so it does not represent every topic or query distribution. • The current questions were used to improve the ground truth, rubric, and evaluation code. They are best treated as a development benchmark rather than an independent final test. • The same gpt-5.4-mini model answered and judged at temperature 0.2. No second judge or formal human inter-rater agreement was measured. • Only vector Top K = 7 and one hybrid weighting were included in the final controlled comparison. The results do not establish the best setting for every query type. • The quality index assigns PARTIAL a value of 0.5 by design. It is useful for summarizing results but should always be read with the raw outcome rates and per-question evidence. • The 361 result rows are repeated measurements of the same 19 questions, so they should not be treated as 361 independent observations or used to claim formal statistical significance.

Specific improvements or next steps
Response: **Improvements to Evaluation Framework**

© MIT xPRO® 2026. All rights reserved.

- Freeze GT01-GT19 as the development set and create a new held-out set before claiming general performance improvement. Create a larger library of questions for development, testing and validation. Validation should include articles and subjects that are not covered by development/testing.
- Use a different, stronger judge at a deterministic setting and manually double-score a sample to estimate judge agreement.
- Record judge rationales and separate factual correctness, completeness, source support, quotation fidelity, and instruction compliance into distinct diagnostic metrics.
- Add retrieval precision, context tokens, latency, and API cost to the report.
- Add controlled paraphrases, no-answer cases, misleading context, conflicting sources, and prompt-injection attempts.
- Use vector retrieval as the current primary baseline, while testing whether query classification and routing can apply BM25 selectively where exact terms help.
- Preserve tables and list structure, and use deterministic Python for joins, counts, rankings, date filters, and correlations after retrieval..

Improvements to Data/Retrieval.

- While sending the entire article content seemed to show better results. (capstone 2.1 chunks vs sections vs full articles). Capstone 3.1 results showing answer quality degradation with 3+ articles. This suggests we need to pre-process data and synthesize better retrieval to find balance between sending complete information needed and reducing noise.

  RAG: html to markdown (vs text) breaking by section (not chunks)
  GraphRAG: multi-article data and long data tables can benefit from introducing graph-like relationships (ex. Nominated vs won actors linked to specific Award years)

Supporting Materials

Use this section to list or link to any supporting materials included with your submission.

Supporting material	Location, link, or note
wikipedia_GT.csv	../GT/wikipedia_GT.csv - 19 questions with reference answers, grading notes, designated sources, scope, evaluation role, and complexity. Submitted with the worksheet.

© MIT xPRO® 2026. All rights reserved.

AI

Supporting material	Location, link, or note
Metric report and experiment outputs	ragas_evaluation_report_checkpoint3_1.pdf - submitted visual report. Raw run CSVs are under ragas_experiments/experiments/.
Evaluation and analysis code	capstone_checkpoint_3_1_oracle_AI_only.py; capstone_checkpoint_3_1_with_retriever.py; ../utils/analyze_ragas_experiments.py.
Ground-truth audit and historical evidence	../GT/capstone2.1_gt_cleanup_notes.md and ../Capstone Checkpoint 2.1/wikipedia Q-A GT.xlsx.
Framework validation evidence	validate_framework() in both Checkpoint 3.1 scripts, plus the GT07 and GT10 examples documented in this worksheet and the raw experiment CSVs.

Assistance Disclosure

I used ChatGPT to help audit and restructure the ground-truth questions, organize the evaluation code, develop the PASS/PARTIAL/FAIL rubric, diagnose measurement problems, and create the experiment-analysis report.

I also used ChatGPT to help organize my observations and refine this written analysis.

I designed the tests to be performed, I reviewed the questions, designed grading rules, experiment outputs, I built evaluation code based on Lab 3.1, reviewed code for ragas report, synthesized final conclusions and made the final decisions about what to include in this worksheet.

End of Required Capstone Checkpoint 3.1

Return to the assignment page in the learning platform to upload and submit your completed capstone checkpoint and complete any remaining items in this module.

© MIT xPRO® 2026. All rights reserved.